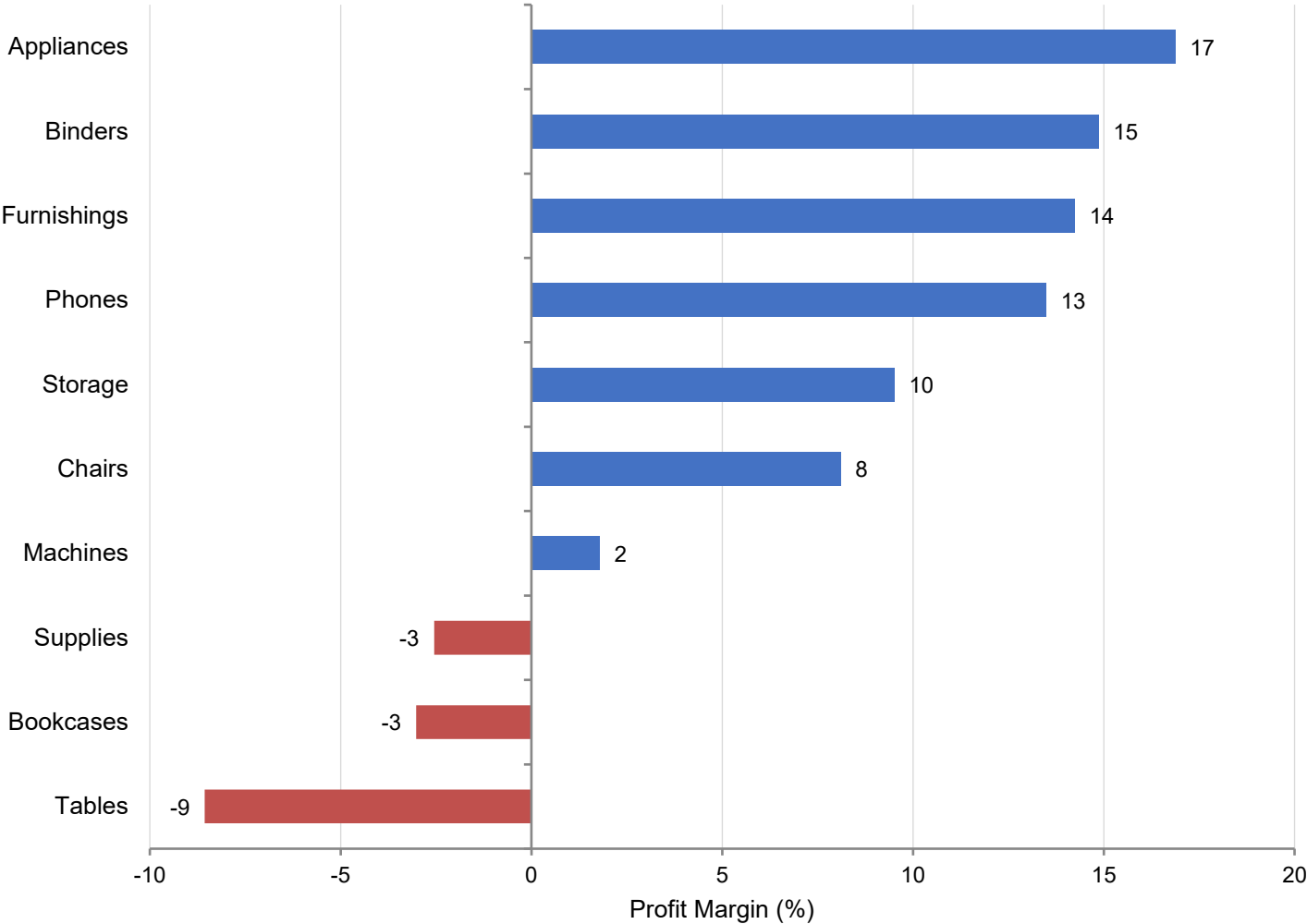

Superstore Sales Analysis

Where sales convert into profit, and where they quietly don't

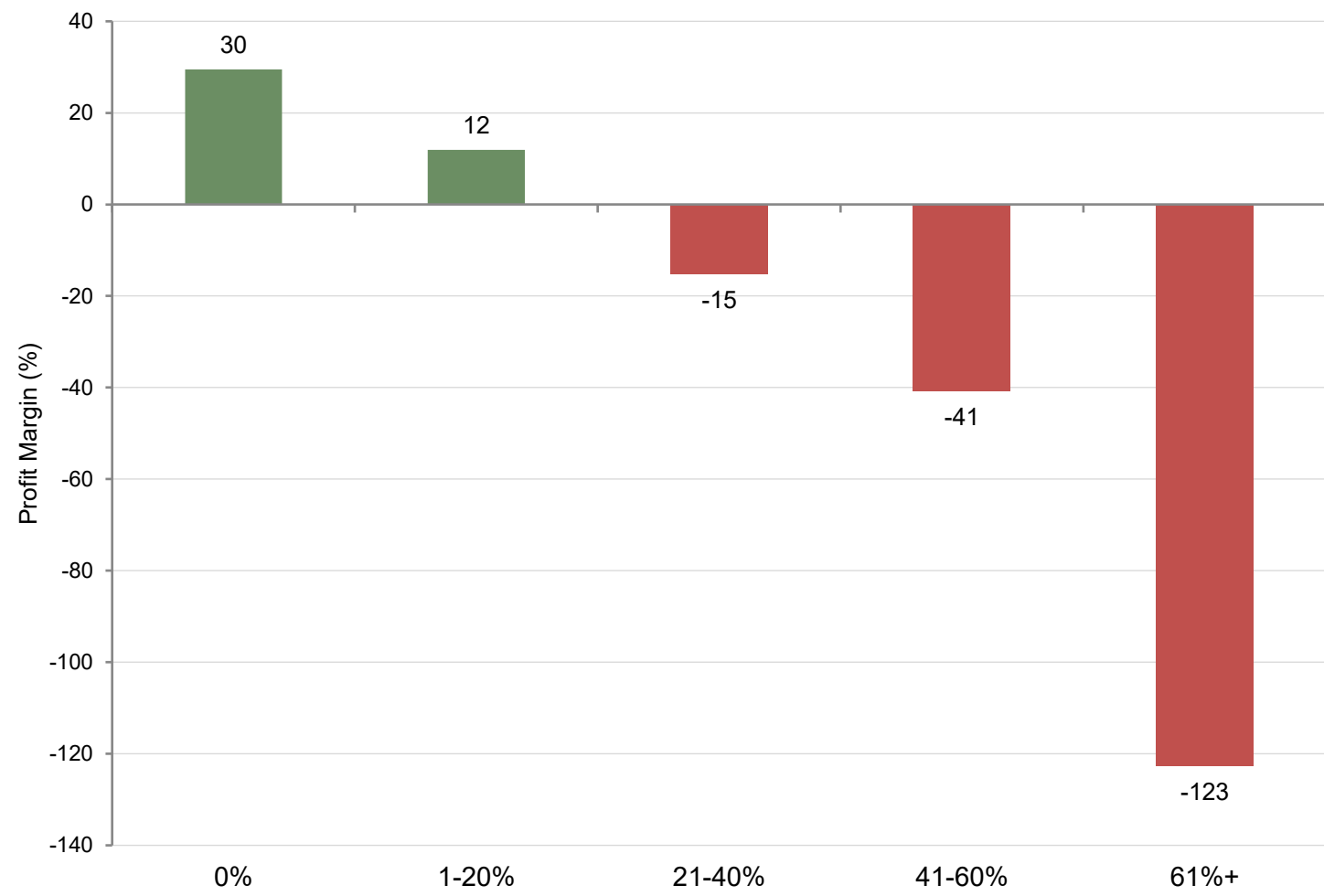
Tables, Bookcases and Supplies are the only sub-categories losing money outright



Tables (-8.56%), Bookcases (-3.02%) and Supplies (-2.55%) are the only three sub-categories with negative overall profit margin.

Tables and Bookcases also rank among Furniture's best-selling products by revenue, so higher sales volume in these two is actively increasing losses, not offsetting them. Supplies sits inside Office Supplies, a category that looks healthy overall, so its loss is easy to miss without breaking down to sub-category level.

Profit margin turns negative once discount passes roughly 20%



At no discount, margin sits at 29.5%.

It turns negative from 21-40% discount, and reaches -122.6% at 61%+, meaning the store loses more than the sale is worth.

A 20% discount cap is the clearest single policy lever available in this data.

Discount alone doesn't fully explain why some products lose money

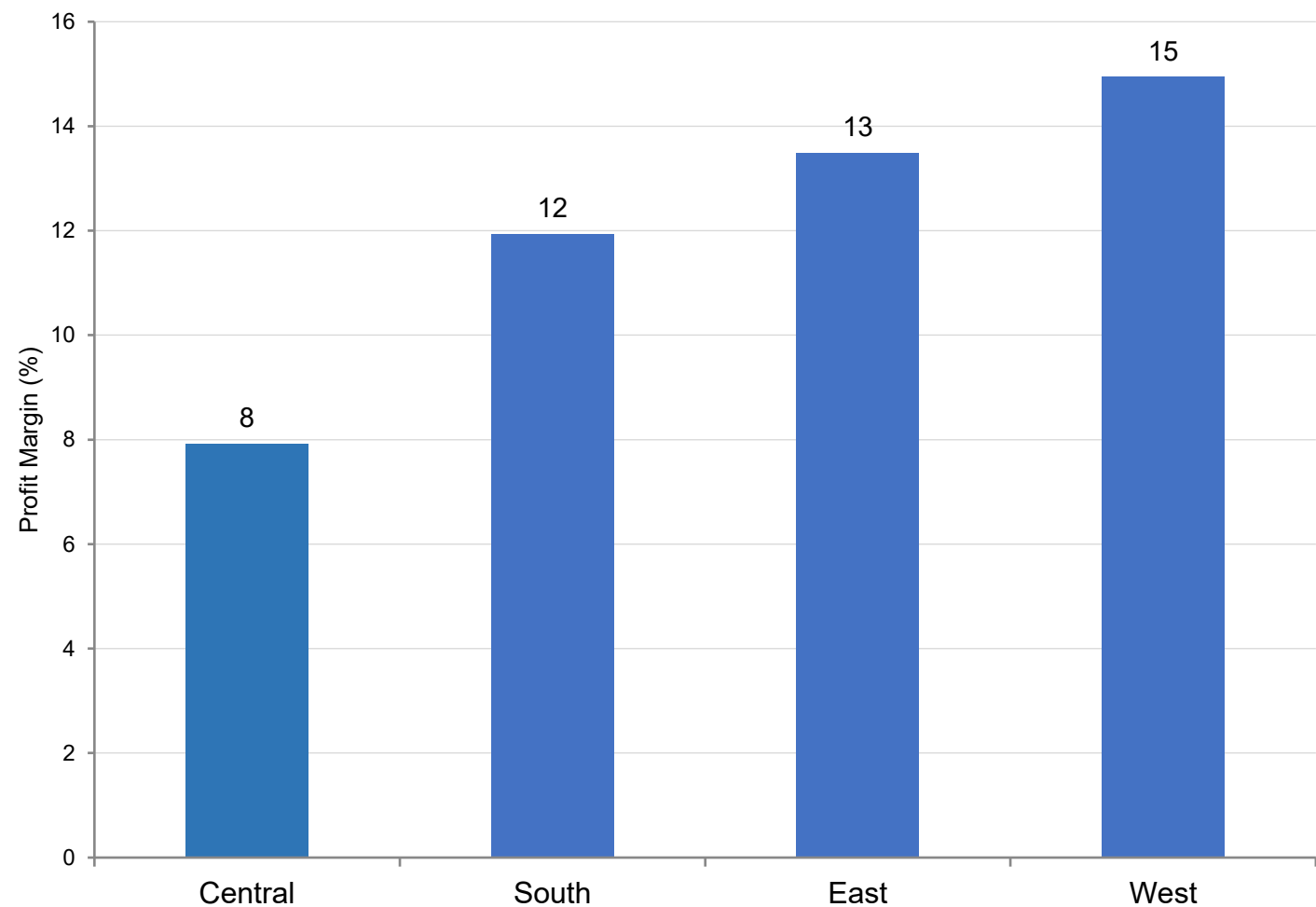
Same discount level, very different outcome

Sub-Category	Avg. Discount	Profit Margin
Binders	37.2%	+14.9%
Tables	26.1%	-8.6%

Binders is discounted more heavily than Tables (37.2% vs 26.1% average) but stays comfortably profitable, while Tables loses money.

Discount is the strongest single driver of profitability in this dataset, but it isn't acting alone here, something specific to Tables and Bookcases (most likely product cost or markup, which isn't available in this data) compounds the effect.

Central region converts sales to profit less efficiently than its peers



Central's margin (7.9%) is roughly half of West's (14.9%), despite similar sales volume to East.

If Central matched West's margin on its existing sales, that's close to double its current profit, without needing any additional sales.

A predictive model flags likely loss-making orders before they happen

91%

Precision

of orders flagged as loss-making genuinely are

77%

Recall

of true loss-making orders are correctly caught

Discount

Top Driver

accounts for over 80% of the model's predictions

The model predicts whether a proposed order will be profitable using sales value, discount, quantity, product type, region and segment, none of which require the order to have already happened.

This means it can be run at the quoting stage, before a discount is confirmed, flagging likely loss-making combinations upfront rather than only discovering the problem after the sale.

Recommendations

1 Cap discount at 20%

This is consistently the point where margin turns negative across the dataset. A formal approval threshold above this level would prevent most avoidable losses.

2 Review Tables, Bookcases & Supplies pricing

These are the only three sub-categories losing money outright. Confirm whether this is deliberate stock clearance or a genuine pricing problem.

3 Investigate Central region's margin gap

Margin is roughly half of West's on similar sales. Check whether heavier regional discounting or product mix is driving the gap.

4 Use the model on proposed orders

Since none of its inputs require the order to have already happened, run it at the quoting stage to flag likely loss-making orders before they're confirmed.

Thank you

Full SQL analysis, model build, and code available in the accompanying notebook.